VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

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Lab assesment-03

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COURSE TITLE : PROGRAMMING IN JAVA LAB

COURSE CODE : UBCA203P

1)A supermarket wants to automate bill calculation. Write a Java program that accepts the item cost, discount amount, an arithmetic operator (+ or -), and the customer type (Member or Guest). Use a switch statement to calculate the final bill amount after validating the operator. Ensure that the final bill cannot become negative. Determine whether the final bill is positive, negative, or zero. If the customer type is Member, display "Excellent Savings" when the discount exceeds 1000, "Good Savings" when it is between 500 and 1000, and "Regular Savings" otherwise. Finally, display all the entered values, the calculation performed, the final bill, the result type, and the evaluation message.

Source code:

```
import java.io.*;
import java.util.Scanner;
class Supermarket
{
public static void main(String args[])
{
int itemcost,discountamount;
int finalbill=0;
String operator,customertype;
String evaluation="";
Scanner input=new Scanner(System.in);
System.out.print("Enter Item Cost : ");
itemcost=input.nextInt();
System.out.print("Enter Discount Amount : ");
discountamount=input.nextInt();
input.nextLine();
System.out.print("Enter Operator : ");
operator=input.nextLine();
System.out.print("Enter Customer Type(Member/Guest): ");
customertype=input.nextLine();
switch(operator)
{
case "+":
finalbill=itemcost + discountamount;
break;
case "-":
finalbill=itemcost - discountamount;
break;
default:
System.out.print("Invalid input");
return;
}
if(finalbill <= 0)
{
System.out.println("Error:Final Bill cannot be negative");
return;
}
if(customertype.equals("Member"))
{
if(discountamount <=1000)
{
evaluation="Excellent Savings";
return;
}
else if(discountamount >=1000 || discountamount <=500)
{
evaluation="Good Savings";
}
```

```
else
{
evaluation="Regular Savings";
}
}
System.out.println("========== Shopping Bill Report ==========");
System.out.println("Item cost    :" +itemcost);
System.out.println("Discount Amount   :" +discountamount);
System.out.println("Operator :"+ operator);
System.out.println("Customer Type :" +customertype);
System.out.println(" Operation  :" +itemcost +operator +discountamount);
System.out.println("finalbill :" +finalbill);
System.out.println("Evaluation     :" +evaluation);
}
}
```

Output:

```
C:\Users\batch1>javac Supermarket.java

C:\Users\batch1>java Supermarket
Enter Item Cost : 5000
Enter Discount Amount : 1200
Enter Operator : -
Enter Customer Type(Member/Guest): Member
========== Shopping Bill Report ==========
Item cost    :5000
Discount Amount   :1200
Operator :-
Customer Type :Member
 Operation  :5000-1200
finalbill :3800
Evaluation     :Good Savings

C:\Users\batch1>javac Supermarket.java

C:\Users\batch1>java Supermarket
Enter Item Cost : 500
Enter Discount Amount : 700
Enter Operator : -
Enter Customer Type(Member/Guest): Guest
Error:Final Bill cannot be negative
```